

UNIT 9

Surveying

Unit Overview

9.1 Introduction

9.2 Clinometers

9.3 Prismatic Compass

Learning Objectives

- *Understand the importance of surveying in general.*
- *Measuring the height of an object using clinometers.*
- *Conduct and calculate the prismatic compass survey of a given spot.*

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9.1 Introduction

Surveying has been essential element in the development of man's environment for centuries. It is important in planning and execution of every form of construction. Surveying is used to map the earth above and below the sea, prepare navigational map and establish databases for natural resource management, development of engineering data for huge buildings constructions, settlements, roads, railways, bridge constructions and so forth.

Surveying is the science of measuring and recording distances, angles, heights and sizes of the earth's surface to obtain data from which accurate plans and maps are made. Generally, surveying is the systematic process of making measurements on the field from which maps are drawn. The map is the most essential piece of equipment which the geographer, engineer and architect use. The geographers must learn to read, make and use map as an essential element. In this lesson, we will see about clinometers and Prismatic Compass.

9.2 Clinometers

In this Instruction, we will learn how to use clinometers to measure the height of a tall object. What you will need; Clinometers, Tape, Paper, Pen or pencil, Assistant.

Step 1: Pick a Spot

Let us pick a spot to measure your object which may be a tree or electric or telephone pole. You should be far enough away from your object that you can see the top of it, and you need to be on level ground with the base of the object.

Step 2: Measure Angle

Here's where we bust out our handy clinometers. Look through the straw of your clinometers at the top of the tree or whatever object you're measuring. The weighted

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